

AMJED MOHAMMED

Senior Software Engineer | AI Applications, LLM Agents & RAG | .NET, Angular, LangGraph

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PROFESSIONAL SUMMARY

Senior Software Engineer and Solutions Architect with 10+ years of experience designing and delivering enterprise-grade web platforms, SaaS products, and AI-integrated systems. Specialized in ASP.NET Core, Angular, Flutter, and microservices architecture, with a proven track record of building GRC platforms, LMS/CRM systems, and government-integrated solutions for the Saudi market. Leads projects end-to-end from architecture and development through deployment, with hands-on expertise in multi-cloud infrastructure (Azure, Digital Ocean, Alibaba), AI/LLM integration (RAG, vector databases, vLLM), and multi-tenant SaaS design. Independent product owner and developer of multiple commercial platforms in production.

PROFESSIONAL EXPERIENCE

Senior Software Engineer & Solutions Architect - AI Applications | Management Solutions Systems Co. (MSSC)

09/2025 – Present

Riyadh, KSA • mssc.com.sa

- Architected and developed an enterprise Learning Management System (LMS) integrated with Saudi Jadarat (competency) — the national skills and certifications— enabling automated trainee enrollment, competency tracking, and certification issuance.
- Designed and built a unified CRM + LMS platform for HSTC (Horizon Solutions for Training), consolidating student lifecycle management, course delivery, training operations, and customer relationships in a single multi-tenant solution.
- Integrated AI and LLM capabilities into the platform for automated content processing, semantic search, intelligent assessment generation, and adaptive learning recommendations.
- Acted as solution architect, defining technical strategy, system design, and integration patterns with Saudi government services and third-party APIs.
- Led the migration of legacy modules to a modern ASP.NET Core + Angular stack with improved scalability, observability, and deployment automation.
- Collaborated with business stakeholders to translate training and accreditation requirements into production-ready features aligned with Saudi regulatory standards.

Senior Software Developer | [HakTrak](#)

07/2023 – 09/2025

- Architected and delivered Comply360, a full-stack GRC platform, using ASP.NET Core Web API and Angular 18 with real-time dashboards powered by SignalR.
- Optimized system performance by integrating Redis caching for compliance reports and session management, reducing database query response times by 60%.
- Implemented background job processing with Hangfire for automated compliance scans, vendor risk assessments, and scheduled SecurityScorecard data imports.
- Containerized applications using Docker (multi-stage builds) and established CI/CD pipelines across Digital Ocean, Alibaba Cloud, and Hetzner environments.
- Led SecurityScorecard API integration using .NET HttpClient with retry and circuit-breaker policies for resilient external service communication.
- Configured SendGrid email services for automated compliance notifications, risk alerts, and system-generated reports with delivery tracking.

- Established testing strategies (xUnit for .NET, Jasmine/Karma for Angular) achieving 90%+ code coverage, and mentored junior developers in modern engineering practices.

Senior Full Stack Developer | [Kafaaat](#)

06/2021 – 06/2023

- Architected microservices-based solutions in .NET Core, each handling a single business capability and communicating via lightweight HTTP APIs and messaging.
- Implemented asynchronous, event-driven communication using RabbitMQ to decouple services and ensure reliable, scalable workflows.
- Built payment integration with Moyasar and global platform integrations with WhatsApp, LinkedIn, Nafath, Wathq, Twitter, and Google — each encapsulated as an independent microservice.
- Containerized services with Docker and orchestrated them on Azure Kubernetes Service (AKS); designed CI/CD pipelines in Azure DevOps for automated builds, tests, and deployments.
- Monitored production health via Azure Monitor and centralized logging using the ELK stack (Elasticsearch, Logstash, Kibana).
- Developed responsive Angular front-ends consuming microservices APIs and authored unit, integration, and contract tests to validate cross-service workflows.

Senior Full Stack Developer | [Oasis Computer Systems](#)

04/2020 – 06/2021

- Designed and delivered two insurance web aggregator platforms (Applus and Almamoon Brokers) using ASP.NET Core, Angular, MS SQL Server, and Redis.
- Implemented real-time integration with multiple insurance company APIs to submit user quote requests and process returned offers in real time.
- Integrated the HyperPay payment gateway for secure, end-to-end transaction processing and reconciliation.
- Architected reusable Angular components and services, focusing on performance, reusability, and maintainability across both platforms.

Full Stack Developer | [LemonLabs for Technical Solutions](#)

08/2017 – 03/2020

- Led a team of 15+ technical support specialists, providing coaching, performance reviews, and career development.
- Monitored the production environment to identify, document, and escalate errors and bugs in collaboration with the development team for rapid resolution.
- Managed and optimized the ticketing system — logging, categorizing, prioritizing, and tracking support requests to ensure SLA compliance.
- Validated new features in staging environments and oversaw configuration management in production, applying updates and patches following best practices.

Software Engineer | [Insight for Multi Activities](#)

12/2014 – 08/2017

- Developed and maintained a Diagnostic Clinic Management System handling patient appointments, medical records, and doctor–patient workflows using C#, VB, SQL Server, and ASP.NET.
- Built a Pharmacy Management System for prescription creation, inventory tracking, and dispensing workflows, integrated with the clinic system.
- Developed a Laboratory Information System managing test ordering, result reporting, approval processes, and automated notifications.

SELECTED INDEPENDENT & CLIENT PROJECTS

Enterprise GRC Platform | [thubut.co](#)

Independent product • Sole Architect & Developer • In active development

- End-to-end Governance, Risk, and Compliance platform built on a Framework-as-Data model with deep closure-table hierarchies and a unified Governance + Risk + Compliance integration centered on a ControllInstance core.

- Stack: PostgreSQL 16, ASP.NET Core 10, Angular 20, Flutter 3, Qdrant, Redis, MinIO, Keycloak.
- AI layer: vLLM (Qwen 2.5), BGE-M3 embeddings, Surya OCR for document understanding, control-mapping, and semantic search.
- Aligned with Saudi PDPL, NCA, SDAIA, DGA, and MCIT regulatory frameworks; targeted as a commercial product for the Saudi GRC market.

KFSHRC Conferences & Events Platform | motamarat.kfshrc.edu.sa

Client: King Faisal Specialist Hospital & Research Centre • Development Team Member

- Contributed as part of the development team building the official conferences and events management platform for KFSHRC, one of the largest healthcare institutions in the Middle East.
- Worked on registration workflows, event content management, and integrations across the conference lifecycle.

Amaaken (أماكن) — Venue Booking Platform | amaaken.sa

Sole Developer • End-to-end architecture and implementation • Pre-launch

- Designed and developed a complete venue booking SaaS platform from scratch — backend, frontend, mobile, and infrastructure — covering venue listings, availability calendars, booking flows, payment processing, and host/guest dashboards.
- Built using ASP.NET Core, Angular, and Flutter with a multi-tenant architecture ready for the Saudi market.

TECHNICAL SKILLS

Backend: ASP.NET Core, C#, .NET 8, Web API, MVC, Microservices, Hangfire, SignalR, RabbitMQ, gRPC

AI / LLM: LLM Agents, Agentic Workflows, LangGraph, Prompt Engineering, Tool Calling, Structured Outputs, Guardrails

RAG & Retrieval: RAG Pipelines, Qdrant, BGE-M3, Vector/Hybrid Search, Chunking, Grounding, vLLM, Evaluation, OCR

Frontend: Angular (9 - 20), TypeScript, RxJS, NgRx, Tailwind CSS, Bootstrap, Material

Mobile: Flutter, Dart, Cross-platform development

Databases: PostgreSQL, MySQL, MS SQL Server, MongoDB, Redis, Entity Framework Core

Architecture: Microservices, Multi-tenant SaaS, Clean Architecture, CQRS

DevOps & Cloud: Docker, Azure DevOps, GitHub Actions, GitLab CI, Coolify, Nginx

Cloud Platforms: Microsoft Azure, Digital Ocean, Alibaba Cloud (KSA), Hetzner, AWS

KSA Integrations: Nafath, Wathq, Moyasar, HyperPay, SADAD

Identity & Security: Keycloak, OAuth 2.0, JWT, OpenID Connect, Role-Based Access Control

Testing: xUnit, NUnit, Moq, Jasmine, Karma, Jest, Integration & Contract Testing

Methodologies: Scrum, Agile, Code Reviews, Mentoring, Cross-functional Leadership

EDUCATION

Bachelor of Science in Engineering

09/2009 – 07/2014

Faculty of Engineering, University of Science and Technology — Khartoum, Sudan

LANGUAGES

Arabic (Native) **English** (Professional working proficiency)